

Segmentation Homework #3

The data for the homework can be found in tutorial3.zip file.

1. Bayer Pattern conversion

In subdirectory „color segmentation“ you can find a raw image obtained from a camera with RGGB Bayer pattern as bayer.ppm. Write a Matlab script that converts this image into RGB image and display it on the screen. Compare it with the correct result from color.ppm

2. HSI color space

Write a Matlab script to convert an RGB image to HSI space (you can use rgb2hsi() Matlab function if you do not success in this for the second step. Use this function to segment red regions in the image „california.jpg“. Make sure to handle the areas with low saturation correctly. Explain, what happens in this case.

3. Facedetector

Build a facedetector for the images provided in directory „Skin Detector“. It provides training images consisting of Face_Test_x.jpg and a corresponding mask for face information as Face_ref_x.png that can be used to train the detector boundaries for a good segmentation of the face hue variation. Write a Matlab script that finds the faces in the test images.